

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 05 · LECTURE NOTES

Routing and Congestion

The canonical application, and the paradox worth building.

MODULE OUTCOMES

- Model routing as a congestion game.
- Explain potential functions and what they guarantee.
- Construct and explain Braess's paradox.
- Explain tolls and other interventions.
- Apply the framework to a computing system.

1 The model

A CONGESTION GAME IS AGENTS CHOOSING RESOURCE SETS, WHERE EACH RESOURCE'S COST DEPENDS ONLY ON HOW MANY CHOSE IT

Each agent picks a subset of resources — a path through a network, a set of machines, a set of links — **and each resource's cost is a function of its load, identical for everyone using it.**

Which covers road traffic, network routing, load balancing, shared caches, bandwidth allocation, and essentially any contended resource — **the structure is identical and only the vocabulary differs**, which is what makes this the canonical application of the whole field.

And the key property is that an agent's cost depends on the others only through the counts — **not on who they are** — **which is exactly what makes the potential function argument work (§2) and is the condition to check before applying any of this.**

So the model is narrow enough to prove strong things about and wide enough to apply to real systems — **which is an unusually good position for a model to be in**, and is why congestion games get a module of their own rather than an example.

2 Potential functions

$$\Phi(\mathbf{s}) = \sum_r \sum_{i=1}^{\text{load}(r)} c_r(i)$$

- **Sum, over each resource, of that resource's cost at each load level from 1 up to its current load** — **a single global function of the entire strategy profile**, due to Rosenthal.

- **And any single agent's improving deviation decreases Φ by exactly that agent's own gain — which is the magic of the construction: a selfish improvement and a decrease in the global potential are the same event, so the two perspectives coincide move by move.**
- **So pure Nash equilibria exist, and best-response dynamics converge to one: Φ takes finitely many values and strictly decreases at every step, so the process must terminate — at a local minimum of Φ , which is precisely a pure equilibrium.**
- **Which is Module 03 §4's second tractable case, and it is why congestion games are the setting in which equilibrium analysis is actually usable rather than merely definable.**

AND THE POTENTIAL IS NOT THE SOCIAL COST, WHICH IS EXACTLY WHERE THE INEFFICIENCY LIVES

Φ sums each resource's cost over every load level up to the current one; the social cost is the load multiplied by the cost at the current load — two different functions of the same profile, and they are only equal for constant cost functions.

So minimising Φ , which is what the dynamics actually do, is not minimising social cost — and the discrepancy between them is precisely the price of anarchy (Module 04), which gives that ratio a mechanical explanation.

Which gives the right intuition for the inefficiency: an agent joining a resource pays the current cost and ignores the increase it thereby imposes on everybody already using it — the externality — and sums of private costs are not the social cost.

And that observation is immediately the design fix (§4): charge each agent the externality it imposes, and selfish choice becomes socially optimal — which is one of the cleanest results in the subject.

3 Braess's paradox

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THE NETWORK, one unit of traffic from s to t
s -> v : latency x      v -> t : latency 1
s -> w : latency 1      w -> t : latency x

equilibrium: half take s-v-t, half take s-w-t
each pays  $1/2 + 1 = 3/2$ 

NOW ADD a zero-latency link v -> w
s -> v -> w -> t has latency  $x + 0 + x$ 
for any single agent this is better than  $3/2$ 
so EVERYONE switches
new equilibrium: all use s-v-w-t, with cost
 $1 + 0 + 1 = 2$ 

EVERYBODY IS WORSE OFF ( $2 > 3/2$ ) AFTER ADDING
CAPACITY. No one behaved irrationally; the new route
really is better for each agent given what the others
are doing.

AND IT IS NOT A CURIOSITY: removing roads has
improved traffic flow in real cities.

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Nobody behaved irrationally — the new route is genuinely better for each individual agent given what the others are doing, at every point in the process, **and the result is worse for all of them**. That is the whole content of the paradox, and it is why 'the agents should just coordinate' is not an analysis. **Project 1 asks you to construct and verify this**, because building it is considerably more convincing than reading it.

4 Interventions

- **Marginal cost tolls** — charge each agent the externality it imposes on the others, and the resulting equilibrium is socially optimal — which is clean, provable, and requires knowing the cost functions.
- **Capacity**: add it carefully — §3 shows that adding capacity can make things worse, so network changes need checking rather than assuming, which is an unusual engineering conclusion and a correct one.
- **Restrict the strategy space** — remove the problematic route entirely — which is crude, effective, and is what cities do when they close a road.
- **Stackelberg routing** — centrally control some fraction of the traffic and let the remainder choose freely — which improves the equilibrium with only partial control, and is directly applicable when you own some of the clients.
- **And doing nothing**, when the price of anarchy is small (Module 04 §4) — which is frequently the right answer and is the one an engineer is least likely to propose. **Marginal cost pricing requires knowing the cost functions** — and in a computing system you frequently do, which makes this far more applicable here than in road traffic.

AND THE COMPUTING APPLICATIONS ARE DIRECT

Load balancers, content delivery routing, inter-domain routing, shared caches, database connection pools, and thread pools — all congestion games, and all exhibiting the same structural inefficiency once clients choose for themselves.

And the externality framing is the useful one: a client retrying aggressively imposes cost on everybody else and bears only a fraction of it — which is exactly why retry storms happen, and why they are not fixed by asking clients to be considerate.

So exponential backoff, admission control, and rate limiting are all externality pricing in disguise — and seeing them that way suggests when they will and will not work: they work when the price tracks the externality, and fail when it does not (a fixed rate limit under-prices at low load and over-prices at high).

Which is the real value of this model for this program — not the traffic examples, but recognising the structure in systems you build (Module 01 §2's table, and CSCE 678's distributed systems material, where these mechanisms appear without the game-theoretic vocabulary).

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Rosenthal — A class of games possessing pure-strategy Nash equilibria	§2's potential function , in the original — three pages.
Braess (1968), translated (free)	§3 — the paradox in its original form, with the translation and commentary.
Roughgarden — Selfish Routing and the Price of Anarchy	§§1 to 4 — the book-length treatment, including tolls and Stackelberg routing. Library copy.
Nisan et al., chapters 18 and 19 (free PDF)	§§2 and 4 — potential games and the interventions, with the proofs.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Model three computing systems** as congestion games.
- 2 **Check the counts-only condition** in each, and find one where it fails.
- 3 **Compute Rosenthal's potential** for a small instance.
- 4 **Verify that an improving move decreases it by the agent's gain.**

- 5 **Run best-response dynamics** and plot the potential.
- 6 **Compare the potential to the social cost** along that path.
- 7 **Construct Braess's paradox** and verify both equilibria.
- 8 **Compute the marginal cost toll** for Pigou's example and check it restores optimality.
- 9 **Implement Stackelberg routing** for a fraction of the traffic.
- 10 **Analyse a retry storm** as an externality problem.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define a congestion game and give the key property.
- 2 Name four computing systems with this structure.
- 3 Give Rosenthal's potential and what it guarantees.
- 4 Why does a selfish improvement decrease it?
- 5 Why is the potential not the social cost?
- 6 State the externality, and what it explains.
- 7 Construct Braess's paradox and give both costs.
- 8 Why is nobody irrational in it?
- 9 Name five interventions and what each requires.
- 10 Why are backoff and rate limiting externality pricing?